OAK INTERNATIONAL SCHOOL

PREMIUM MATHEMATICS ACTIVITY WORKBOOK — CLASS 2

Grade Level: Sixième (6ème) / IB MYP 2 (Ages 11–12)

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Programmes: IB Middle Years Programme (English) & Beninese National APC Curriculum (Français)

SECTION A — STUDENT INTRODUCTION / INTRODUCTION POUR L'ÉLÈVE

Stepping into Secondary Mathematics Excellence

Entrée dans l'excellence mathématique du premier cycle secondaire à Oak International School !

In Sixième / MYP 2, mathematics shifts from simple arithmetic into the wonderful realm of deductive reasoning and formal geometry:

- In **IB MYP 2**, you explore how symmetry and spatial relations govern natural structures and architecture, learning to communicate lines of reasoning (Criterion C) and investigate general algebraic patterns (Criterion B).
- In **Beninese 6ème (APC)**, you discover the statutory power of **Situation d'Apprentissage N°1 (Configurations du plan)** and **N°5 (Symétrie orthogonale)**. You learn the formal structure of geometric proof (*démonstration géométrique*), moving from simple observation to absolute mathematical proof.

This workbook unites both traditions: you will master precise French axiomatic phrasing alongside versatile English mathematical inquiry.

SECTION B — BILINGUAL MATHEMATICAL LANGUAGE / GLOSSAIRE BILINGUE

Planar Geometry, Symmetry & Number Theory

English Term	French Equivalent (*Terme Français*)	Mathematical Definition & Symbolic Example	Common Translation Pitfalls
Straight line	*Droite*	An infinite one-dimensional set of points: (d) or (AB).	Do not confuse with "line segment" (*segment*) [AB] which has finite length.
Ray	*Demi-droite*	A portion of a line starting at endpoint A and extending infinitely through B: [AB).	Note bracket convention: square bracket at closed origin [, parenthesis at infinite end).
Perpendicular bisector	*Médiatrice d'un segment*	The line perpendicular to a segment at its midpoint: (\Delta) \perp [AB] at I with IA = IB.	False friend: do not confuse with *médiane* (median) or *bissectrice* (angle bisector).
Axial symmetry / Reflection	*Symétrie axiale / orthogonale*	Transformation across line (d) where (d) is the perpendicular bisector of [MM']: S_{(d)}(M) = M'.	In French, specifically designated *symétrie orthogonale par rapport à une droite*.
Collinear points	*Points alignés*	Three or more points that lie on the same straight line: A, B, C \in (d).	In French, always say *« les points sont alignés »*, not "collinéaires".
Prime number	*Nombre premier*	An integer p > 1 whose only positive divisors are 1 and p: 2, 3, 5, 7, 11, 13, \dots.	1 is neither prime nor composite in both languages!
Highest Common Factor (HCF)	*Plus Grand Commun Diviseur (PGCD)*	The greatest integer that divides each of two or more numbers: \text{HCF}(24, 36) = 12.	Note abbreviation inversion: HCF (or GCD) in English vs PGCD in French.
Lowest Common Multiple (LCM)	*Plus Petit Commun Multiple (PPCM)*	The smallest positive integer divisible by both numbers: \text{LCM}(4, 6) = 12.	In French: PPCM.

SECTION C — SHARED CONCEPTUAL FOUNDATIONS

Unit Theme: Perpendicular Bisectors, Circles and Axial Symmetry

1. The Perpendicular Bisector: Dual Definitions

In both curricula, the perpendicular bisector (\Delta) of a segment [AB] possesses two equivalent definitions:

1. **Geometric Definition**: The straight line perpendicular to segment [AB] passing through its midpoint I.
2. **Locus of Equidistance (Caractérisation par l'équidistance)**: The set of all points M in the plane that are equidistant from the endpoints A and B:

M \in (\Delta) \iff MA = MB

This property is the foundation of circle circumscription: the centre of the circle passing through three non-collinear points is the intersection of their perpendicular bisectors.

2. Axial Symmetry (Orthogonal Reflection) and Invariance

When a plane figure is reflected across an axis of symmetry (d):

- Distances are conserved: A'B' = AB (isometry).
- Alignment is conserved: if A, B, C are collinear, their images A', B', C' are collinear.
- Angles are conserved: \widehat{A'B'C'} = \widehat{ABC}.
- Surface areas are conserved: \text{Area}(F') = \text{Area}(F).

3. Worked Example: Constructing and Proving Symmetries

Problem: Let (d) be a straight line and A a point not on (d). A' is the symmetric image of A with respect to (d). Point M lies on (d). Prove that triangle MAA' is isosceles.

- **The Beninese APC Deductive Proof Format (*Démonstration en 3 étapes*)**:
- *Données (On sait que)* : A' est le symétrique de A par rapport à la droite (d), et le point M appartient à (d).
- *Propriété (Or si)* : La symétrie orthogonale par rapport à une droite conserve les distances, et tout point situé sur l'axe de symétrie est son propre symétrique (S_{(d)}(M) = M).
- *Conclusion (Donc)* : MA = MA', ce qui prouve que le triangle MAA' est un triangle isocèle en M.
- **The IB MYP Reasoning Format (Criterion C)**:
- By definition of reflection across axis (d), (d) is the perpendicular bisector of segment [AA'].
- Since point M lies on the perpendicular bisector (d), it is equidistant from both endpoints: MA = MA'.
- Therefore, triangle MAA' has two equal sides, which satisfies the definition of an isosceles triangle with vertex M.

SECTION D — CONTEXTUALISED BRIDGING ACTIVITIES

Activity: Designing an Eco-Tourism Dock in Ganvié (Lake Nokoué)

1. Problem Situation

The community council of Ganvié (the stilt village on Lake Nokoué) wishes to construct a solar-powered charging dock D along a straight water channel (L) to service two schools of watercraft stationed at locations A and B on the same side of the channel.

To minimise electrical cabling and transit times, the council must determine the exact point D on line (L) such that the sum of distances DA + DB is minimized.

2. The IB Perspective (Inquiry into Geometric Transformation)

- Reflect point B across the channel line (L) to create its mirror image B'.
- For any point X on (L), XB = XB'.
- Therefore, XA + XB = XA + XB'.
- The shortest distance between two points A and B' in a Euclidean plane is a straight line segment [AB'].
- The optimal dock location D is the intersection of line (AB') and channel line (L).

3. The Beninese APC Perspective (Construction & Formal Justification)

- *Construction géométrique* :

1. Construire le point B', symétrique orthogonal du point B par rapport à la droite (L).
2. Tracer la droite (AB').
3. Marquer le point d'intersection D = (AB') \cap (L).

- *Démonstration de minimalité* :
- Soit M un point quelconque de la droite (L) distinct de D.
- Puisque M \in (L) et que (L) est l'axe de symétrie échangeant B et B', on a MB = MB'.
- De même, D \in (L), donc DB = DB'.
- La longueur du trajet passant par D est DA + DB = DA + DB' = AB' (car A, D, B' sont alignés dans cet ordre).
- La longueur du trajet passant par M est MA + MB = MA + MB'.
- D'après l'inégalité triangulaire dans le triangle AMB' : MA + MB' \ge AB' = DA + DB.
- Le point D réalise donc le minimum absolu de la distance totale.

4. Comparison of Methods

- IB guides students through an intuitive spatial transformation and shortest-path principle.
- Beninese APC requires formal construction commands with ruler and compass and an explicit triangular inequality proof.

5. Student Reflection Question

Why does replacing B with its symmetric image B' allow us to transform a bent two-part path (AD + DB) into a single straight line (AB')?

Espace de réponse:

_____

_____

_____
_____

6. Application in a New Context

Two telecom relay antennas P and Q in the Collines department are situated on the same side of the National Interstate Highway RNIE 2 (modelled as line H). Find the location on H for a shared solar generator that minimizes cable installation costs.

Working space:

_____
_____
_____
_____

SECTION E — PROGRESSIVE PRACTICE / EXERCICES PROGRESSIFS

Level 1: Foundational Understanding (Terminology & Divisibility)

1. Complete with the appropriate mathematical symbol (\in, \notin, \perp, \parallel):

a) Point M lies on line (d): M ____ (d)

b) Point P does not belong to segment [AB]: P ____ [AB]

c) Line (d_1) is perpendicular to line (d_2): (d_1) ____ (d_2)

d) Line (L_1) is parallel to line (L_2): (L_1) ____ (L_2)

2. Divisibility criteria: Among the numbers 135, 240, 312, 505, 729:

a) Which are divisible by 2?

b) Which are divisible by 3?

c) Which are divisible by 5?

d) Which are divisible by 9?

Level 2: Standard Application (Constructions & Prime Decomposition)

3. Decompose into a product of prime factors:

a) 72 =

b) 108 =

c) Determine \text{PGCD}(72, 108) and \text{PPCM}(72, 108).

4. On a blank sheet or in your notebook:

a) Draw a segment [AB] of length 6\text{ cm}.

b) Using a compass and straightedge, construct the perpendicular bisector (\Delta) of [AB].

c) Mark a point P on (\Delta) such that IP = 4\text{ cm}, where I is the midpoint of [AB].

d) Measure PA and PB. State their values and verify the equidistance property.

Level 3: Multi-Step Problems (Fractions & Directed Numbers Bridging)

5. Calculate and give the answer in simplest form:

a) \frac{5}{12} \times \frac{18}{25} =

b) \frac{7}{4} \div \frac{21}{8} =

c) \frac{3}{5} + \frac{2}{5} \times \frac{1}{4} = (Remember order of operations!).

6. Bridging Module: Directed Integers:

Evaluate:

a) (+8) + (-5) =

b) (-12) + (-7) =

c) (+6) - (+14) =

d) (-9) - (-15) =

Level 4: Reasoning & Geometric Proof

7. Let [AB] be a line segment with midpoint O. Line (d) is perpendicular to (AB) at O.

Let M be a point on (d) distinct from O.

Let M' be the symmetric image of M with respect to the line (AB).

a) What is the nature of quadrilateral MAM'B?

b) Write a structured geometric proof showing that the four sides of MAM'B have equal lengths.

c) Deduce the specific geometric name of this quadrilateral.

Level 5: Extension & 6ème National Examination Challenge

8. Situation d'Évaluation Type 6ème (SA N°1 & N°5):

Dans la commune de Savalou, un artisan forgeron conçoit une grille de portail décorative.

Sur sa feuille de dessin, il trace une droite (d) et place deux points E et F de part et d'autre de (d) tels que la droite (EF) coupe (d) au point K non perpendiculairement.

Il construit ensuite les points E' et F', symétriques respectifs de E et F par rapport à (d).

1. Fais une figure géométrique soignée respectant les consignes ci-dessus.
2. Démontre que le point K est son propre symétrique par rapport à la droite (d).
3. Démontre que les points E', K, F' sont alignés.
4. Sachant que EF = 8,5\text{ cm}, détermine la longueur du segment [E'F'] en justifiant par la propriété appropriée.

SECTION F — INTEGRATED APPLICATION TASKS

Task F-1: The Traditional Kanvo Weaving Loom (MYP Criterion B & C)

Context: In Abomey, traditional weavers produce *Kanvo* (royal woven fabric) using geometric strip patterns with reflectional symmetries.

A master weaver creates a border design by repeating a symmetrical motif:

- Pattern 1 uses 4 white threads and 1 red central core.
- Pattern 2 uses 8 white threads and 2 red central cores.
- Pattern 3 uses 12 white threads and 3 red central cores.

Questions:

1. Organize the data into a table showing Pattern Number (n), Red Cores (R), and White Threads (W) for n = 1, 2, 3, 4, 5.
2. State the mathematical relationship between the pattern number n and the number of white threads W.
3. Write a general formula for W in terms of n.
4. The weaver uses a spool containing 160 white threads. What is the maximum pattern size n that can be completed?
5. Explain how axial symmetry ensures that the fabric maintains balanced tension along both edges.

SECTION G — PROGRAMME-SPECIFIC REINFORCEMENT

Part G-1: IB MYP 2 Criterion C Communication Task

Write a formal, mathematically rigorous explanation (using full English sentences and standard symbols) explaining why a circle has an infinite number of axes of symmetry, whereas a regular hexagon has exactly six axes of symmetry.

Part G-2: Beninese National Curriculum Geometric Proof

Exercice de Démonstration Formelle

Soit un triangle ABC isocèle en A. Soit (d) la médiatrice de la base [BC].

1. Démontre que le point A appartient à la droite (d).
2. Démontre que la droite (d) est également la bissectrice de l'angle \widehat{BAC}.

Rédige la démonstration selon le modèle officiel : Données / Propriété / Conclusion.

SECTION H — ASSESSMENT, REFLECTION & WORKED SOLUTIONS

Diagnostic Pre-Flight Check

- [ ] I can construct a perpendicular bisector using a compass and straightedge.
- [ ] I know the divisibility rule for 9 (sum of digits is a multiple of 9).
- [ ] I can find the symmetric point of A across a line (d).
- [ ] I can calculate with negative numbers on a number line.

COMPLETE WORKED SOLUTIONS & MARKING SCHEME

Solutions for Section E (Progressive Practice)

1. Mathematical Symbols (1 mark each):

a) $M \in (d)$

b) $P \notin [AB]$

c) $(d_1) \perp (d_2)$

d) $(L_1) \parallel (L_2)$

2. Divisibility Tests (1 mark each):

a) Divisible by 2 (even unit digit): 240, 312.

b) Divisible by 3 (sum of digits multiple of 3): 135 (1+3+5=9), 240 (6), 312 (6), 729 (18). All four!

c) Divisible by 5 (ends in 0 or 5): 135, 240, 505.

d) Divisible by 9 (sum of digits multiple of 9): 135 (9), 729 (18).

3. Prime Factorisation (1 mark each):

a) $72 = 2^3 \times 3^2$.

b) $108 = 2^2 \times 3^3$.

c) $\text{PGCD}(72, 108) = 2^2 \times 3^2 = 4 \times 9 = 36$.

$\text{PPCM}(72, 108) = 2^3 \times 3^3 = 8 \times 27 = 216$. [2 marks]

4. Constructions: Correct verified construction of perpendicular bisector with arcs, $PA = PB = 5\text{ cm}$ (by Pythagoras $\sqrt{3^2+4^2}=5$). [3 marks]

5. Fraction Operations (1 mark each):

a) $\frac{5 \times 18}{12 \times 25} = \frac{1 \times 3}{2 \times 5} = \frac{3}{10}$.

b) $\frac{7}{4} \times \frac{8}{21} = \frac{1 \times 2}{1 \times 3} = \frac{2}{3}$.

c) $\frac{3}{5} + \frac{2}{20} = \frac{3}{5} + \frac{1}{10} = \frac{6+1}{10} = \frac{7}{10}$.

6. Directed Numbers (1 mark each):

a) +3

b) -19

c) -8

d) -9 + 15 = +6

7. Geometric Proof for Quadrilateral MAM'B (4 marks):

- M lies on (d), which is the perpendicular bisector of [AB], so MA = MB.
- M' is the reflection of M across (AB), so (AB) is the perpendicular bisector of [MM']. Hence AM = AM' and BM = BM'.
- Therefore, MA = MB = AM' = BM'.
- A quadrilateral with four equal sides is a rhombus (*losange*).

8. Savalou Artisan 6ème Task (5 marks):

1. Accurate geometric diagram. [1 mark]
2. $K = (EF) \cap (d)$, therefore $K \in (d)$. Any point lying on the axis of symmetry is invariant (its own symmetric image): $S_{(d)}(K) = K$. [1 mark]
3. The reflection of point E is E', the reflection of F is F', and the reflection of K is K. Since E, K, F are collinear (lying on the same line), and orthogonal symmetry preserves alignment, their symmetric images E', K, F' are collinear. [2 marks]
4. Orthogonal symmetry preserves distances. Therefore, $E'F' = EF = 8.5\text{ cm}$. [1 mark]

Solutions for Section F (Task F-1: Kanvo Loom)

1. Table values: n=1 \to W=4; n=2 \to W=8; n=3 \to W=12; n=4 \to W=16; n=5 \to W=20. [2 marks]
2. The number of white threads is directly proportional to n, with a constant factor of 4. [1 mark]
3. Formula: W = 4n. [1 mark]
4. Setting 4n \le 160 \implies n \le 40. The weaver can complete up to Pattern 40. [2 marks]
5. Axial symmetry balances thread tension equally across both sides of the central core, preventing warping or curling of the fabric edges. [2 marks]

Solutions for Section G-2 (Benin Formal Geometric Proof)

- *Données* : Le triangle ABC est isocèle en A, donc AB = AC. (d) est la médiatrice de [BC]. [1 pt]
- *Propriété* : La médiatrice d'un segment est le lieu géométrique des points équidistants des extrémités de ce segment. [1 pt]
- *Conclusion 1* : Comme AB = AC, le point A est équidistant de B et C, donc A appartient à la médiatrice (d). [1 pt]
- *Démonstration bissectrice* : Soit I le milieu de [BC]. La droite (d) passe par A et I. Dans le triangle ABC isocèle en A, la médiatrice de la base est aussi un axe de symétrie du triangle. Par symétrie orthogonale d'axe (d), la demi-droite [AB) a pour image [AC). La symétrie conservant la mesure des angles, elle partage l'angle \widehat{BAC} en deux angles de même mesure: \widehat{BAI} = \widehat{CAI}. La droite (d) est donc la bissectrice de \widehat{BAC}. [2 pts]

End of Workbook 2 (Sixième / MYP 2).